

Physiological Adaptations to Ramadan Fasting Using Consumer Wearable Data: An N-of-1 Longitudinal Study

Abstract

This exploratory N-of-1 longitudinal study investigated the physiological effects of Ramadan fasting using consumer wearable data collected from an Apple Watch and Apple Health. Metrics including Heart Rate Variability (HRV), Resting Heart Rate (RHR) and sleep duration were analyzed across three phases: pre-fasting, fasting, and post-fasting. Daily trends and 7-day trailing averages were visualized to assess autonomic and recovery-related changes over time.

Results demonstrated a decrease in average HRV during the fasting period, followed by a rebound and increase above baseline levels post-fasting. Resting Heart Rate decreased during fasting before returning toward pre-fasting values post-Ramadan. Sleep duration increased during the fasting period, although the relationship between sleep duration and HRV appeared weak. These findings suggest that Ramadan fasting may transiently alter autonomic markers and recovery physiology, with adaptive recovery occurring following the fasting period.

This study highlights the potential of consumer wearable devices for low-cost longitudinal physiological monitoring and exploratory digital health research.

Introduction

Ramadan fasting involves prolonged intermittent fasting from dawn to sunset over approximately one month. In addition to changes in caloric intake and meal timing, Ramadan fasting is often associated with altered sleep schedules, hydration patterns, physical activity timing, and overall circadian rhythm [1, 2].

Heart Rate Variability (HRV) is commonly used as a non-invasive marker of autonomic nervous system activity and recovery status [3]. Higher HRV is generally associated with greater parasympathetic activity and improved physiological recovery, while reductions in HRV may indicate physiological stress, fatigue, or adaptation to environmental and behavioral changes [3, 4]. Resting Heart Rate (RHR) is another commonly used cardiovascular marker that may reflect changes in metabolic demand, autonomic balance, and overall physiological stress [5].

Recent advances in consumer wearables allow for continuous collection of physiological data outside laboratory environments. Devices such as the Apple Watch provide accessible methods for collecting longitudinal cardiovascular and sleep-related data, enabling exploratory analyses of real-world physiological adaptations.

This study aimed to explore how Ramadan fasting affected HRV, RHR, and sleep duration using wearable-derived data collected across pre-fasting, fasting, and post-fasting phases.

Methods

Data Collection

Physiological data were collected using an Apple Watch Series 11 synchronized with Apple Health. Data were exported from Apple Health in XML format and processed into CSV format for analysis. Data from January 2026 onward were included in the analysis. The following variables were analyzed:

- Heart Rate Variability (HRV)
- Resting Heart Rate (RHR)
- Sleep duration
- Sleep stages (i.e. Deep, Core, etc.)

Data Processing

The exported XML file was parsed using Python to extract relevant health metrics. Sleep durations were calculated from start and end timestamps and converted into total duration values. Daily averages were computed for HRV and RHR. Sleep durations were aggregated on a daily basis. A 7-day trailing average was also calculated for HRV to reduce short-term variability and improve trend visualization.

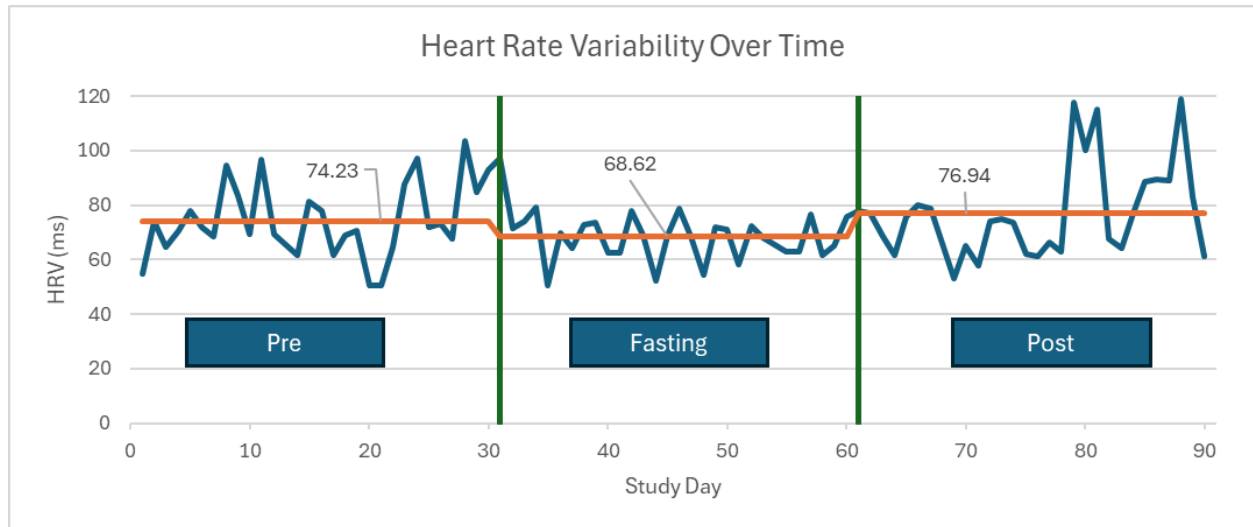
The study period was segmented into three phases:

- Pre-fasting (Day 1 - 30)
- Fasting (Day 31 - 60)
- Post-fasting (Day 61 - 90)

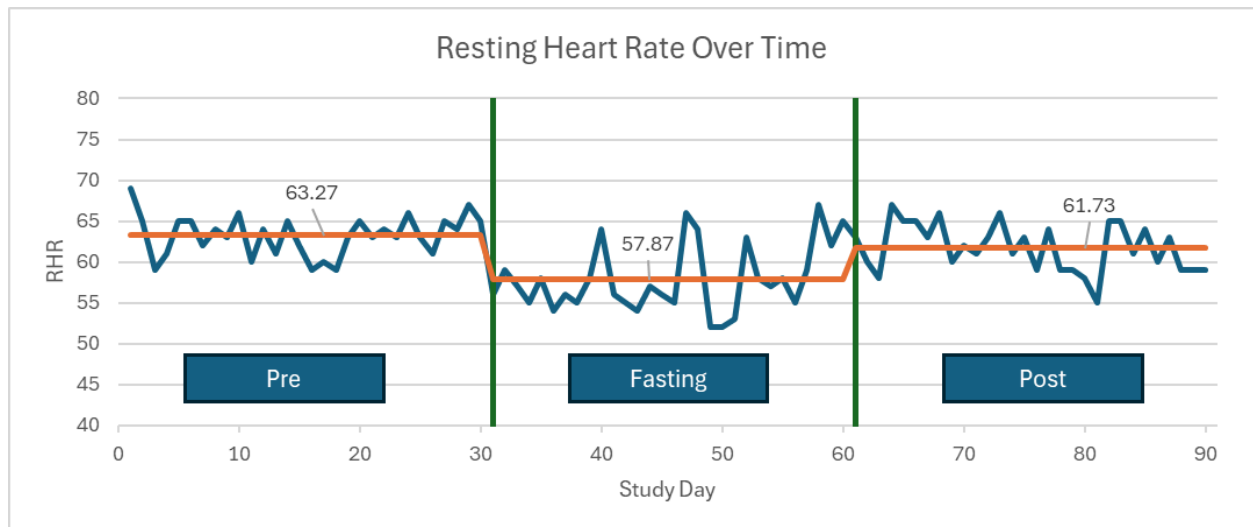
Data Visualization

The following visualizations were created:

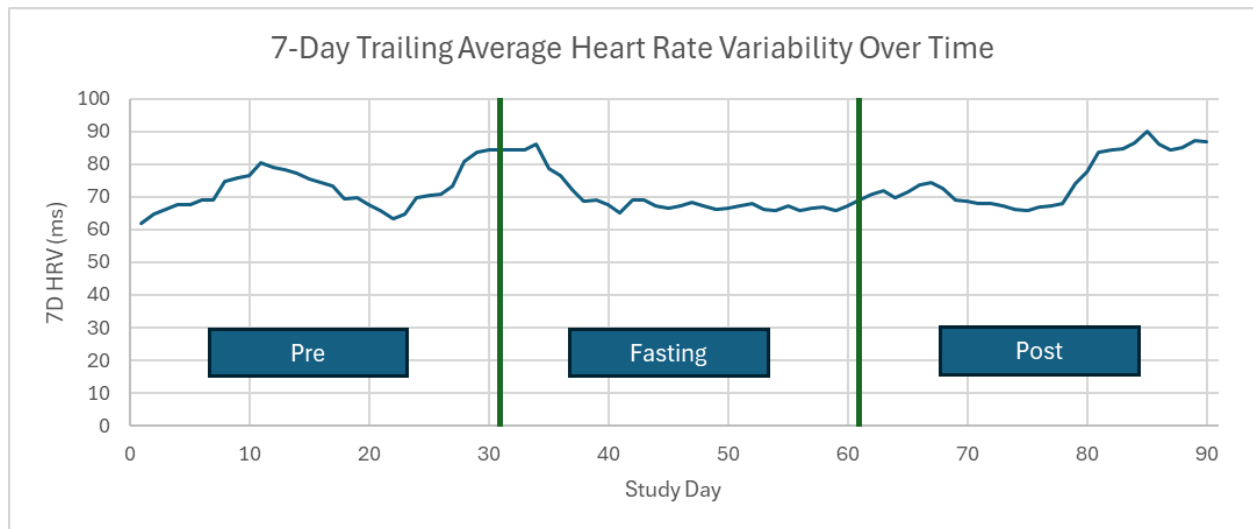
- Figure 1: HRV over time



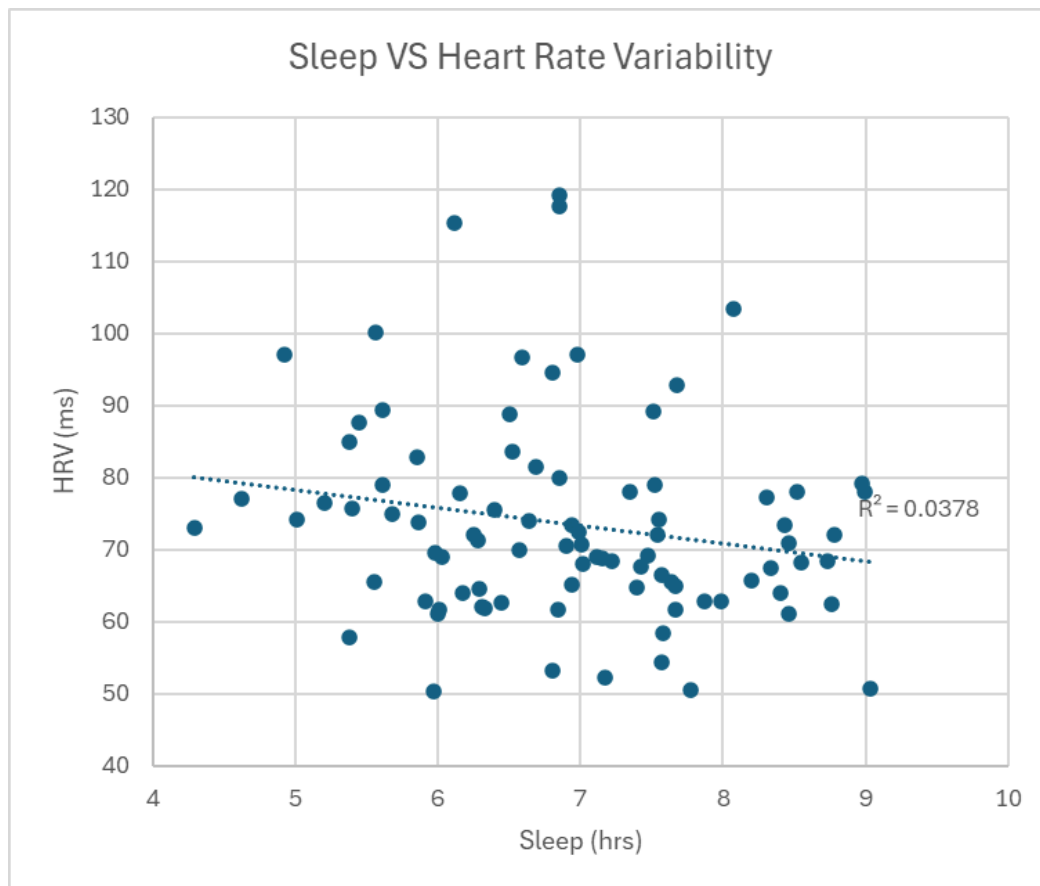
- Figure 2: RHR over time



- Figure 3: 7-day trailing average HRV over time



- Figure 4: Scatter plot of sleep duration versus HRV



All analyses and visualizations were performed using Microsoft Excel.

Results

The physiological data was analyzed across three distinct phases: Pre-Fasting, Fasting, and Post-Fasting. The average values for key metrics in each phase are summarized in Table 1.

Table 1: Summary Statistics by Phase

Phase	Days (n)	Avg HRV (ms)	Avg RHR (bpm)	Avg Sleep (hrs)
Pre	30	74.23	63.27	6.92
Fasting	30	68.62	57.87	7.16
Post	30	76.94	61.73	6.65

Heart Rate Variability (HRV)

As shown in the time-series plot (Figure 1), daily HRV exhibited significant day-to-day fluctuation. The 7-day trailing average (Figure 3) provides a clearer view of the trend, showing a decline in HRV during the fasting period. The average HRV decreased by 7.6% from the Pre-Fasting phase (74.23 ms) to the Fasting phase (68.62 ms). In the Post-Fasting phase, HRV rebounded to an average of 76.94 ms, exceeding the initial baseline.

Resting Heart Rate (RHR)

A distinct downward trend in RHR was observed during the fasting period (Figure 2). The average RHR decreased by 8.5% from 63.27 bpm in the Pre-Fasting phase to 57.87 bpm during the Fasting phase. Following the fast, the average RHR began to return toward baseline, settling at 61.73 bpm in the Post-Fasting phase.

Sleep Duration

Average sleep duration increased by 3.5% during the Fasting phase (7.16 hrs) compared to the Pre-Fasting phase (6.92 hrs). To accurately assess the impact of sleep on recovery, sleep duration for a given night was correlated with the Heart Rate Variability (HRV) recorded the following morning. The scatter plot analysis (Figure 4) revealed no strong linear correlation between these two metrics ($R^2 = 0.0378$). This indicates that within this dataset, total sleep duration alone was not a reliable predictor of autonomic recovery as measured by HRV.

Discussion

This exploratory wearable study identified measurable physiological changes during Ramadan fasting. The observed 8.5% decrease in average RHR during the fasting period aligns with previous research, which has identified fasting periods as being associated with a lower heart rate, likely reflecting reduced metabolic demand and improved cardiovascular efficiency [6].

On the other hand, the 7.6% reduction in HRV observed during the fasting period may reflect transient physiological stress or autonomic adaptation associated with altered meal timing, hydration status, sleep fragmentation, and changes in exercise timing. This finding is noteworthy as it diverges from some studies that have reported an increase or no change in parasympathetic activity during Ramadan [7-9]. Despite the temporary decrease, HRV demonstrated a notable rebound in the post-fasting phase, potentially indicating recovery and physiological adaptation following the fasting period.

Interestingly, while average sleep duration increased by 3.5% during the fasting period, this did not correlate strongly with an improvement in HRV. The very weak correlation ($R^2 = 0.0378$) suggests that sleep quality may be more influential than sleep quantity. This observation is contrary to many large-scale studies that report a decrease in total sleep time during Ramadan, often due to pre-dawn meals (Suhoor) [10, 11]. The fact that increased sleep did not improve HRV in this case strongly implies that other factors such as sleep fragmentation, altered sleep architecture or the timing of caloric intake played a more significant role in modulating autonomic recovery than sleep duration alone.

Ultimately, this N-of-1 study highlights the high degree of individual variability in response to a standardized protocol like Ramadan fasting. While causality cannot be established, the project successfully demonstrates the feasibility of using consumer wearable devices for accessible, longitudinal physiological monitoring and exploratory digital health research.

Limitations

Several limitations should be considered when interpreting these findings:

- This study was limited to a single participant (N=1)
- Consumer wearable HRV measurements may not be equivalent to clinical-grade ECG measurements
- Potential confounding variables such as stress, dietary intake, hydration, and exercise intensity were not strictly controlled
- Sleep quality and sleep fragmentation were not comprehensively analyzed
- The observational nature of the study limits causal interpretation

Despite these limitations, the study provides a practical demonstration of real-world wearable-based physiological monitoring.

Conclusion

This exploratory N-of-1 wearable study suggests that Ramadan fasting may transiently alter autonomic and recovery-related physiological markers. During fasting, HRV decreased while Resting Heart Rate also declined. Following Ramadan, HRV rebounded and exceeded pre-fasting levels, suggesting possible physiological adaptation or recovery effects. The findings also indicate that sleep duration alone may not strongly predict HRV changes, highlighting the multifactorial nature of physiological recovery. This project demonstrates the potential value of consumer wearable devices such as the Apple Watch for conducting accessible longitudinal digital health research and monitoring physiological adaptation in real-world settings.

Future Directions

Potential future extensions of this work include:

- Expanding the study to multiple participants
- Including dietary and hydration tracking
- Incorporating exercise timing and intensity variables
- Applying statistical significance testing
- Using machine learning or predictive modeling approaches for recovery analysis
- Comparing wearable-derived data with clinical-grade physiological measurements

Further studies may help clarify how intermittent fasting influences autonomic recovery and cardiovascular physiology over longer periods.

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